IMPACT: A Toolchain for Nonlinear Model Predictive Control Specification, and Deployment

Overview and application

KU Leuven

February 2025



0 Outline

1. Why do we need IMPACT?
2. What is MPC?
3. What is IMPACT?

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Implementation of **Model Predictive Control (MPC)** in mechatronic systems is a complex task in **modelling**, specifying the strategy and **deploying** it.

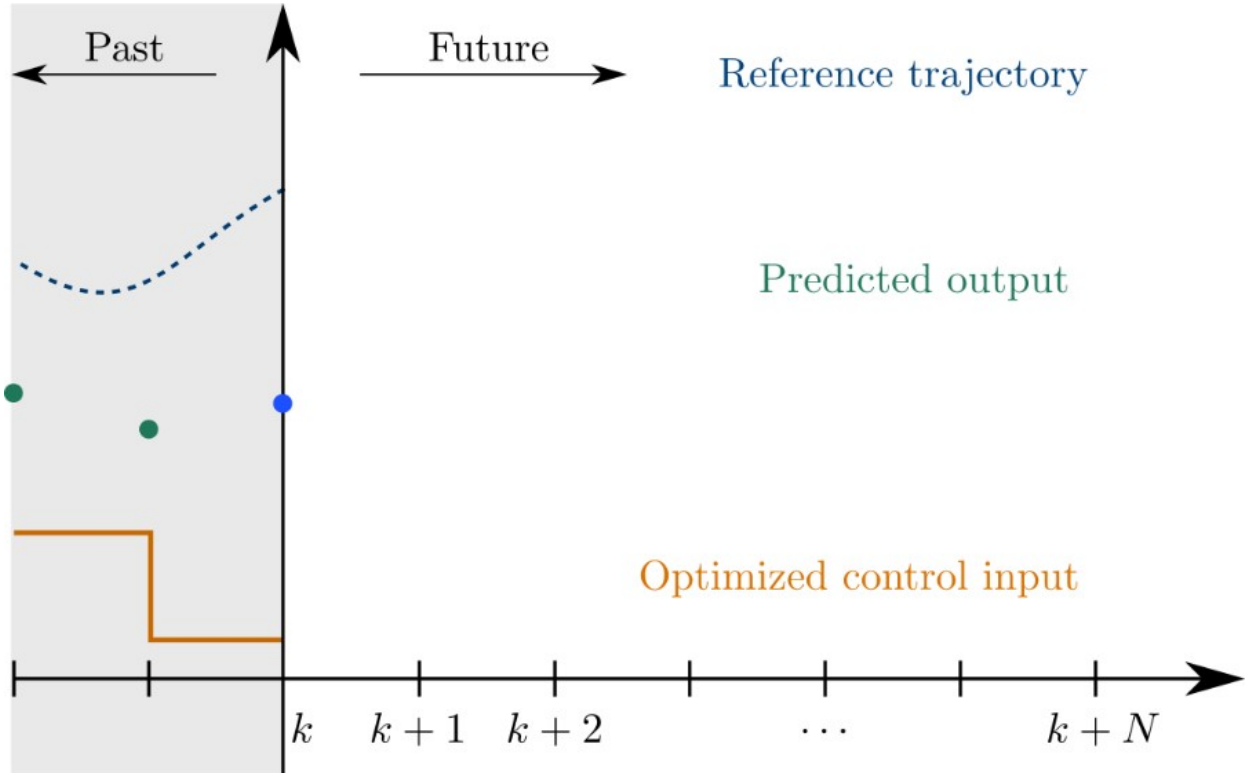
This work presents **IMPACT** toolchain modules which focuses on **facilitating** these tasks

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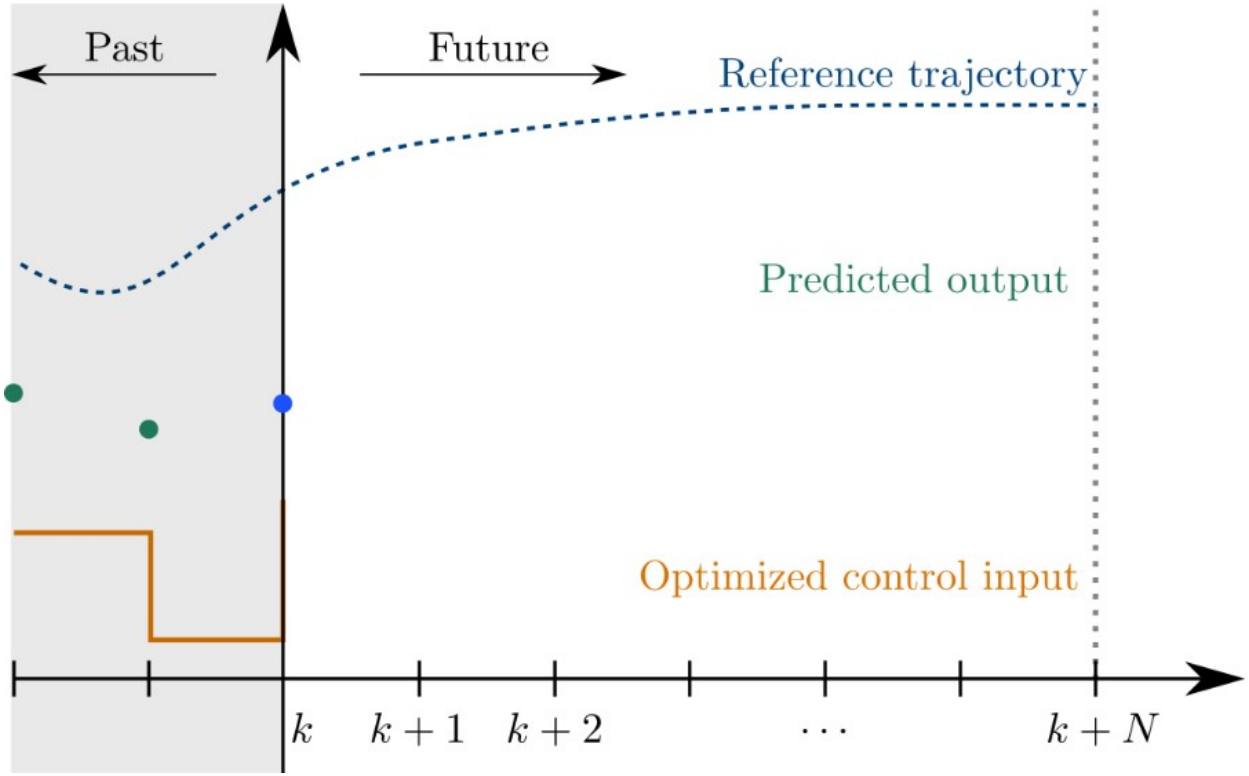
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MPC is a model base technique that optimizes the control action base on the behavior prediction in a time window called horizon.



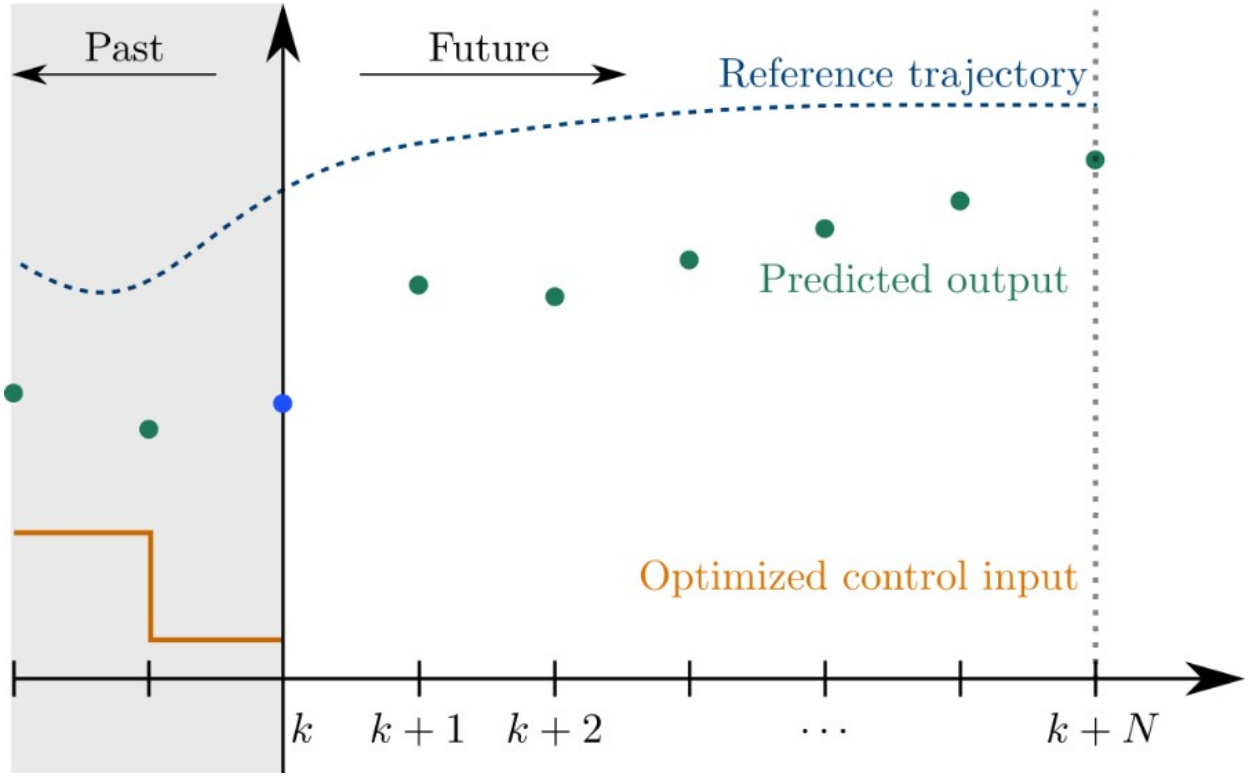
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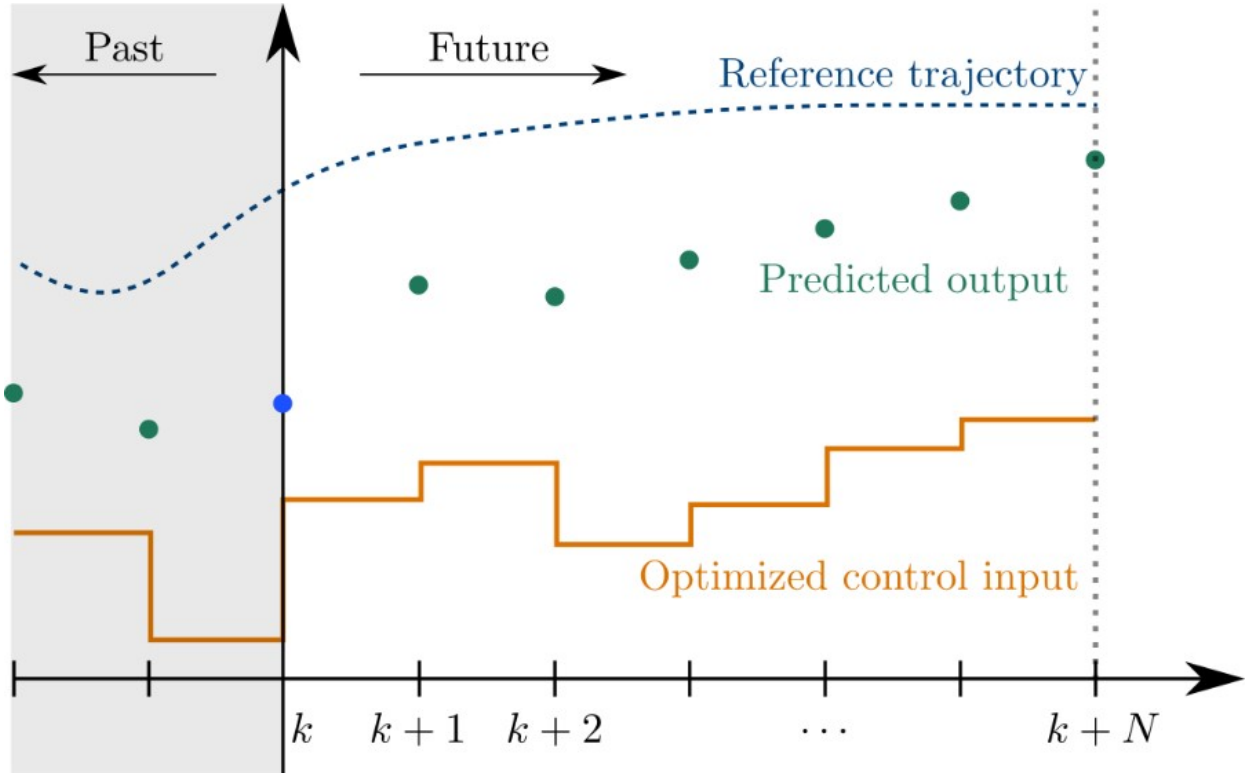
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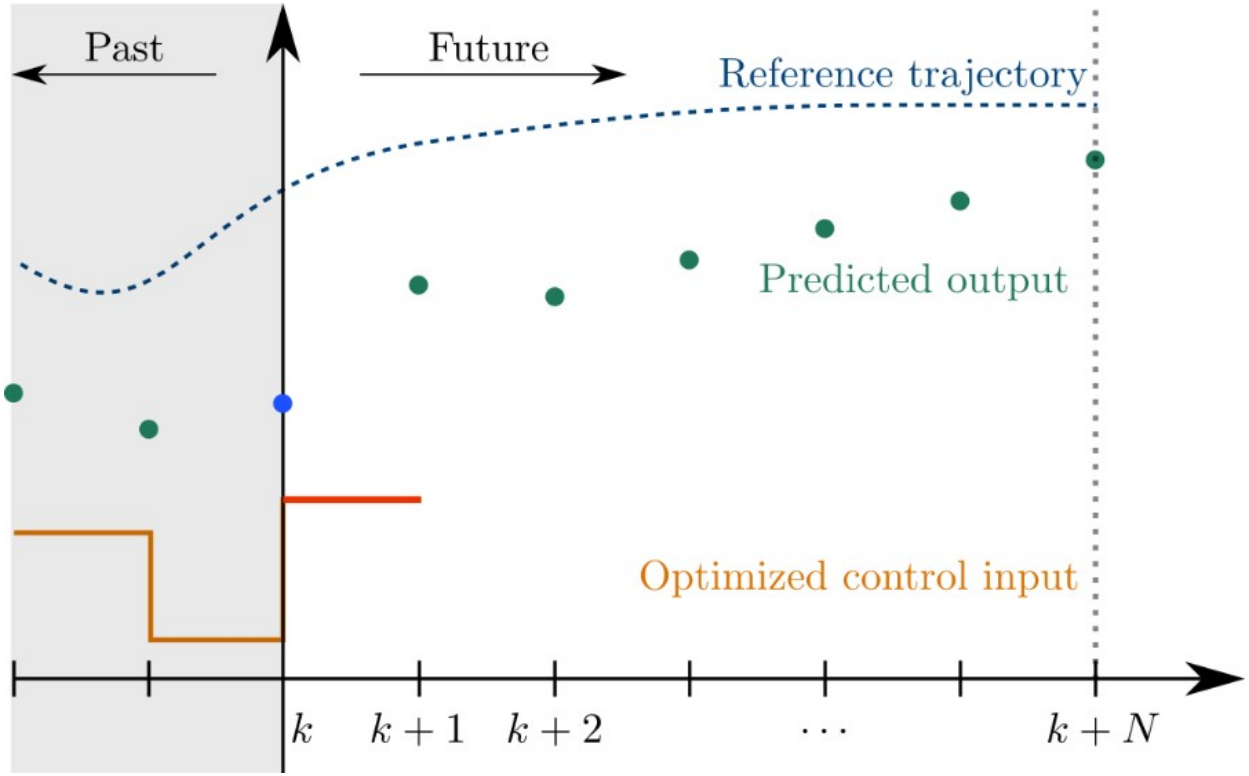
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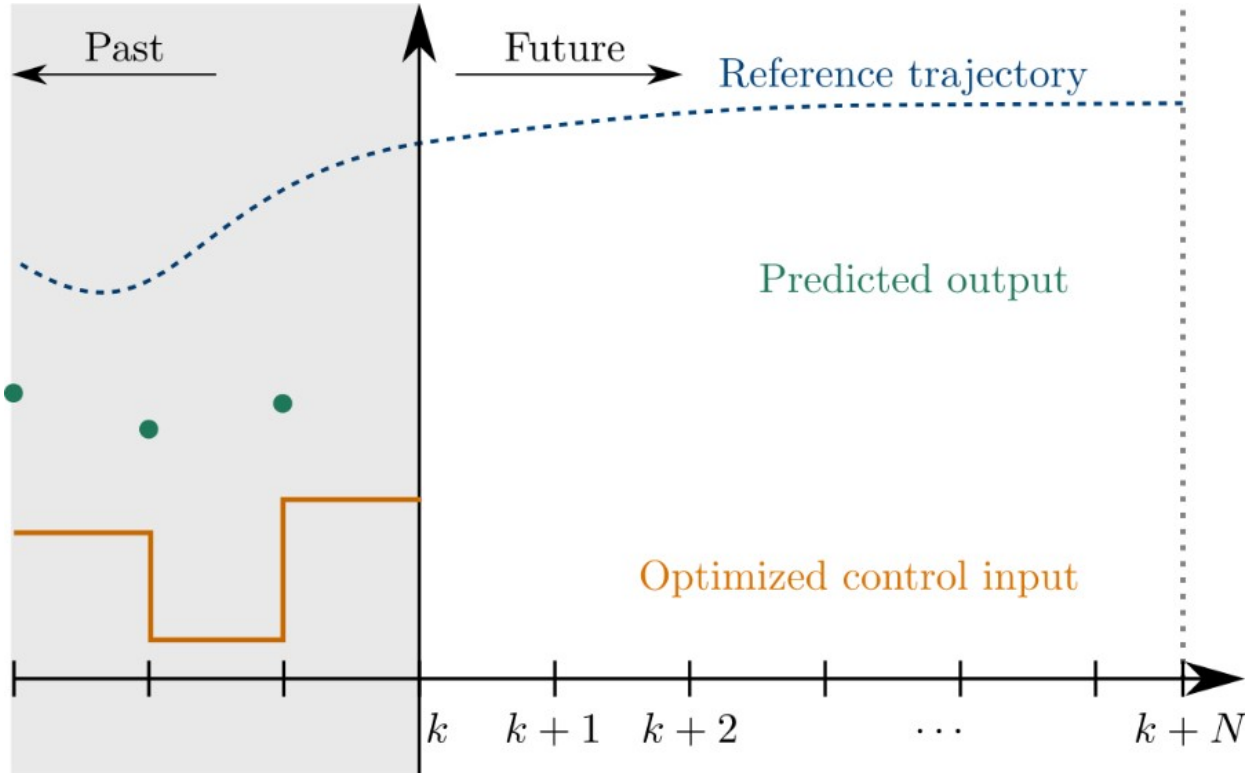
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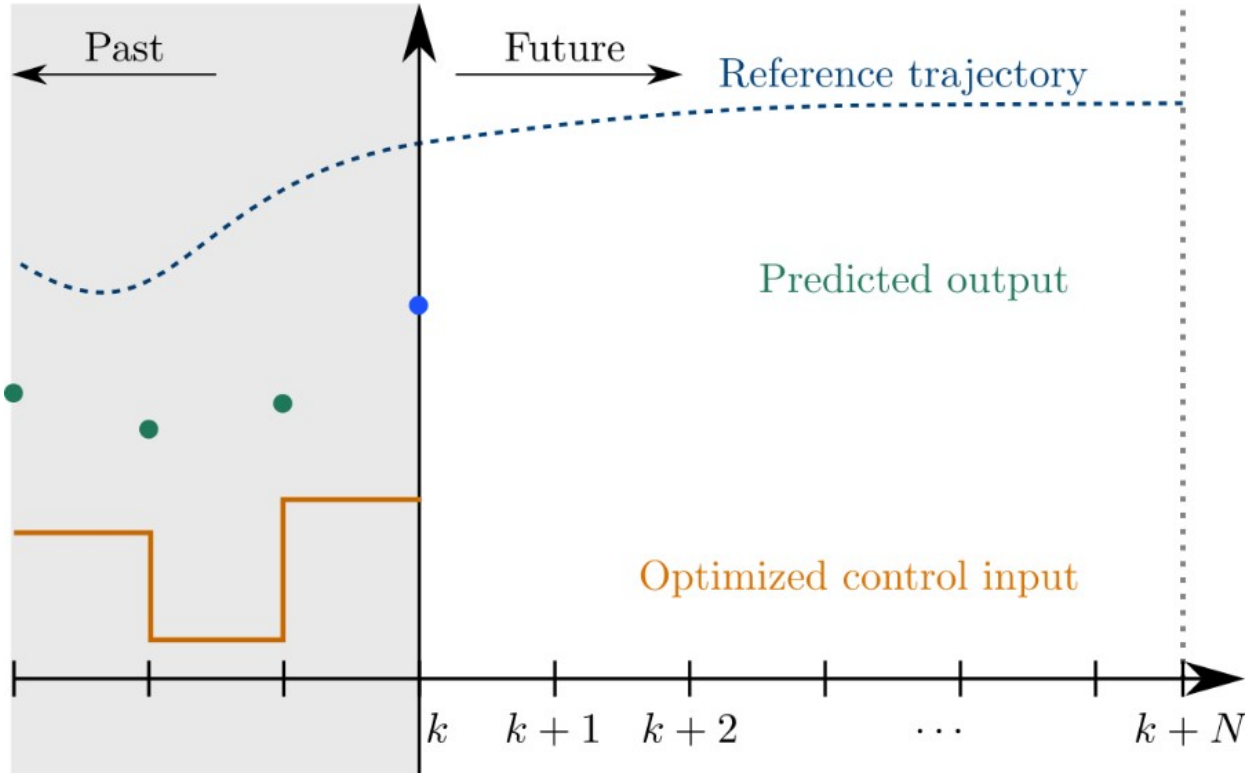
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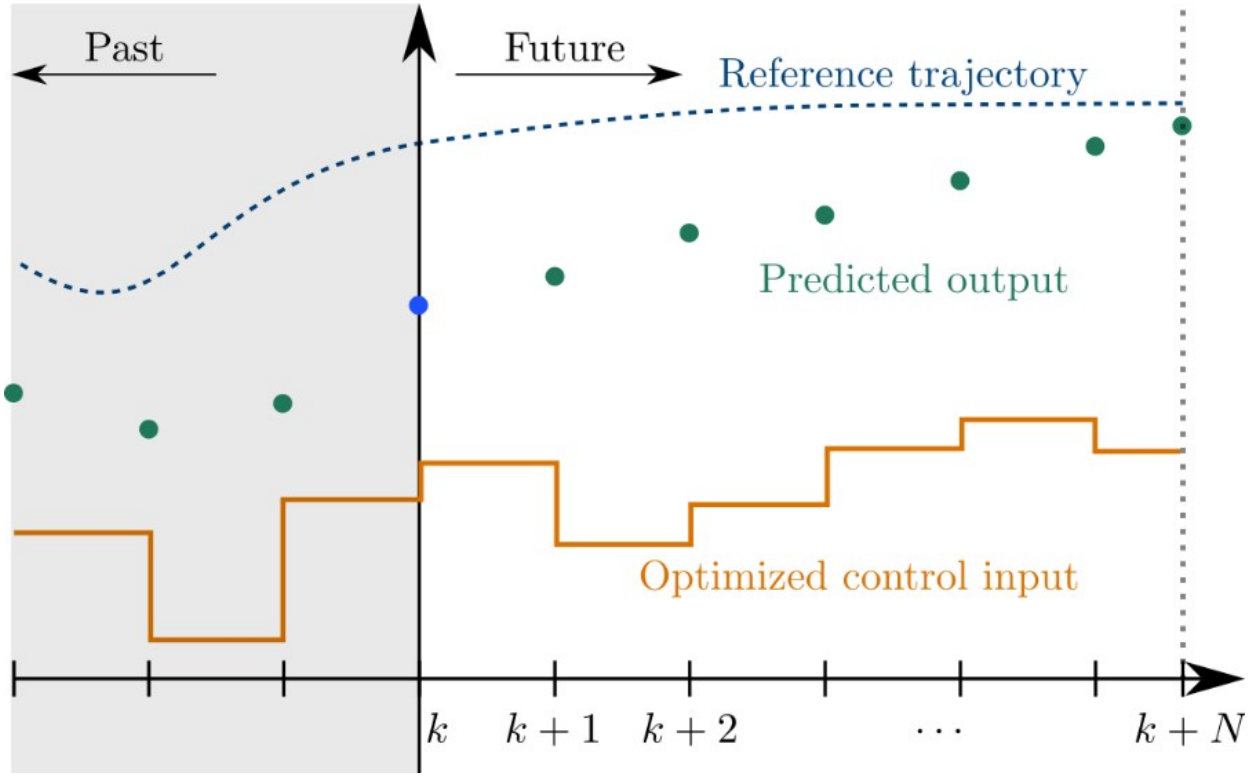
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Considering a dynamic system with a model described by ODE and DAE, where

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The general description of the problem that will be addressed is

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\underset{\boldsymbol{x},\boldsymbol{u},\boldsymbol{z}}{\text{minimize}} \quad & \int_{t_0}^{t_\mathrm{f}} V(\boldsymbol{x}(t),\boldsymbol{u}(t),\boldsymbol{p},t)\mathrm{d}t + V_f(\boldsymbol{x}(t_\mathrm{f}),\boldsymbol{p}) && \text{(1a)}\\
\text{subject to} \quad & B(\boldsymbol{x}(t_0),\boldsymbol{x}(t_f),\boldsymbol{p}) \leq 0, && \text{(1b)}\\
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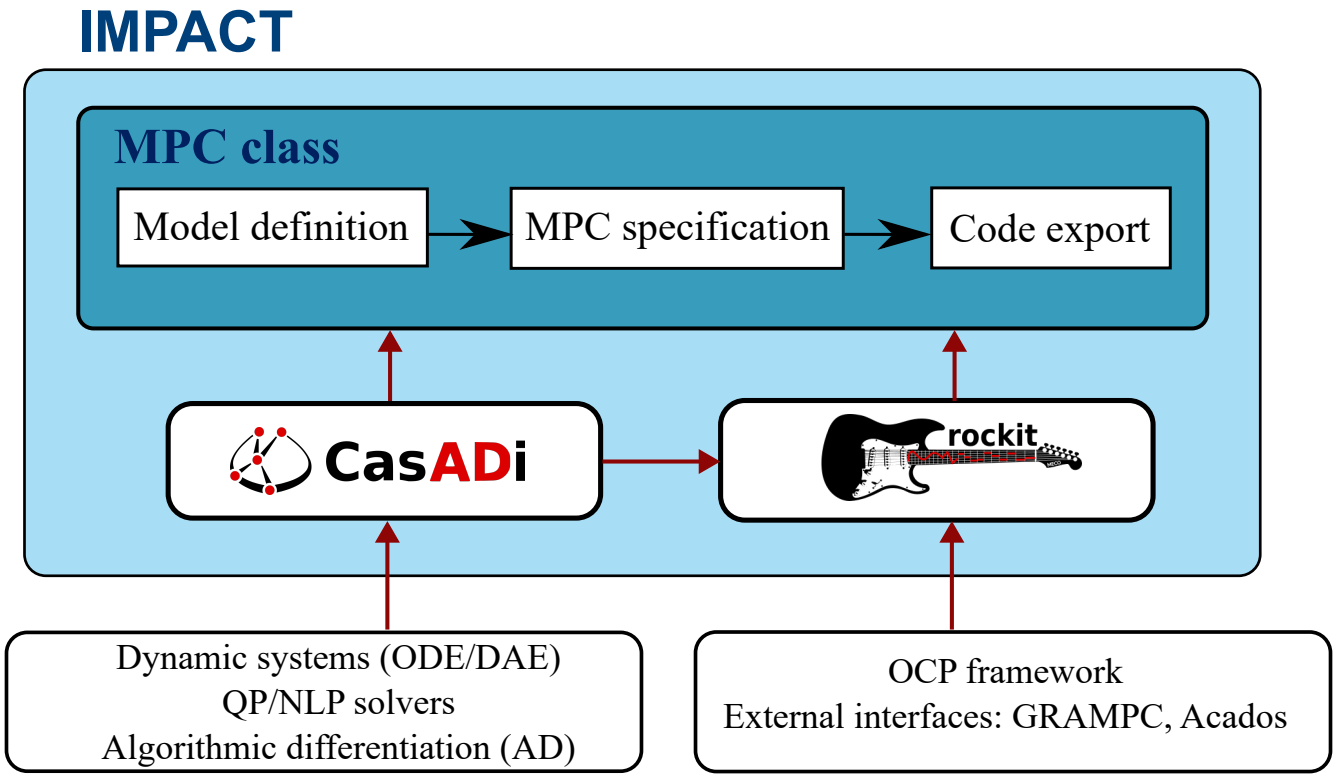
- There is no perfect combination of OCP formulation, transcription method, and optimization solver.
- Proper tools are missing to give the user the flexibility of switching methods easily for fast prototyping.

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3 IMPACT architecture

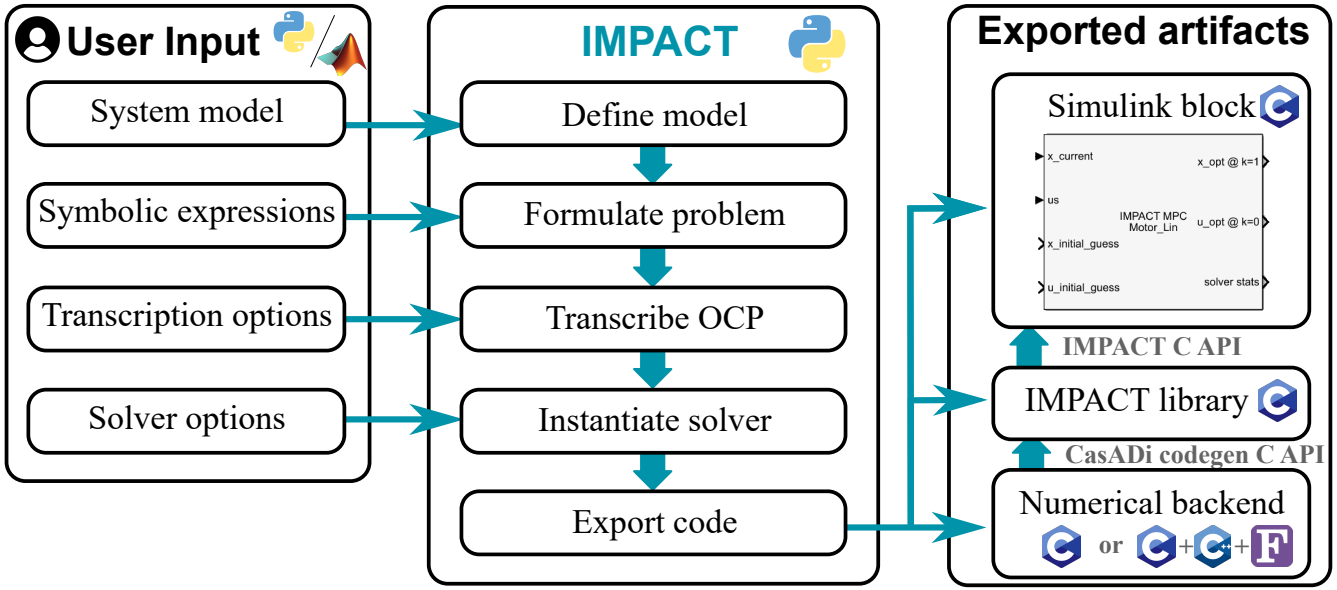
Graphical illustration of IMPACT toolchain structure and its dependencies



All the new features and capabilities of **CasADi** and **Rockit** will be naturally inherited by IMPACT making it easy to keep it on track.

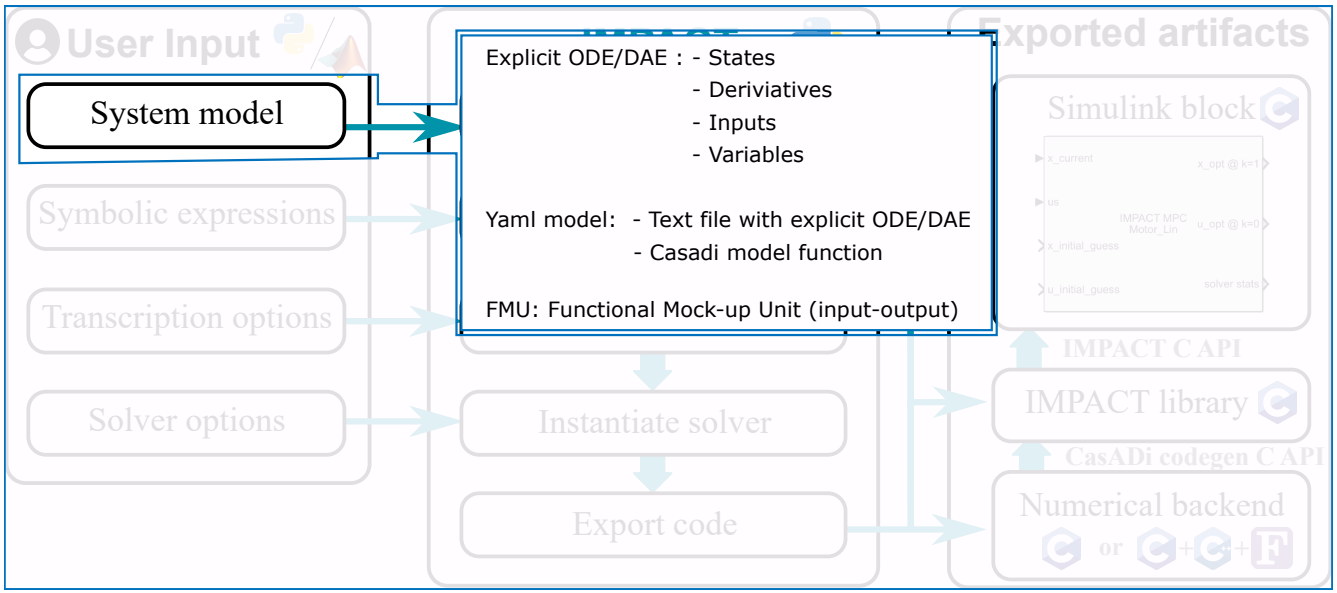
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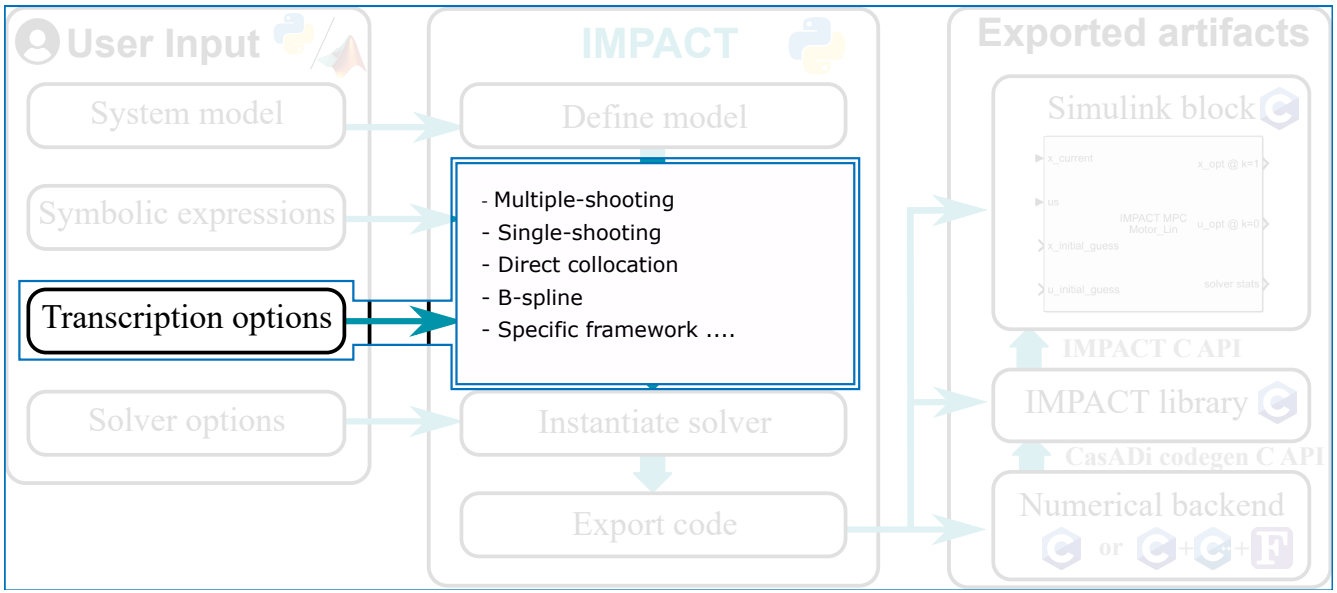
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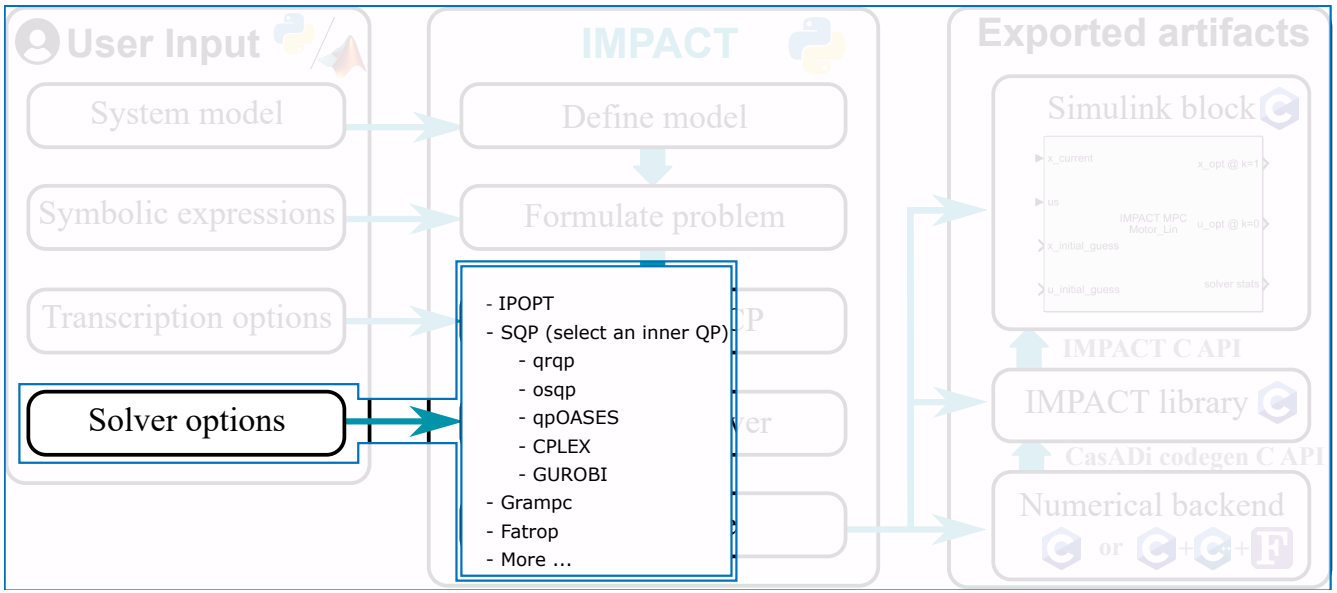
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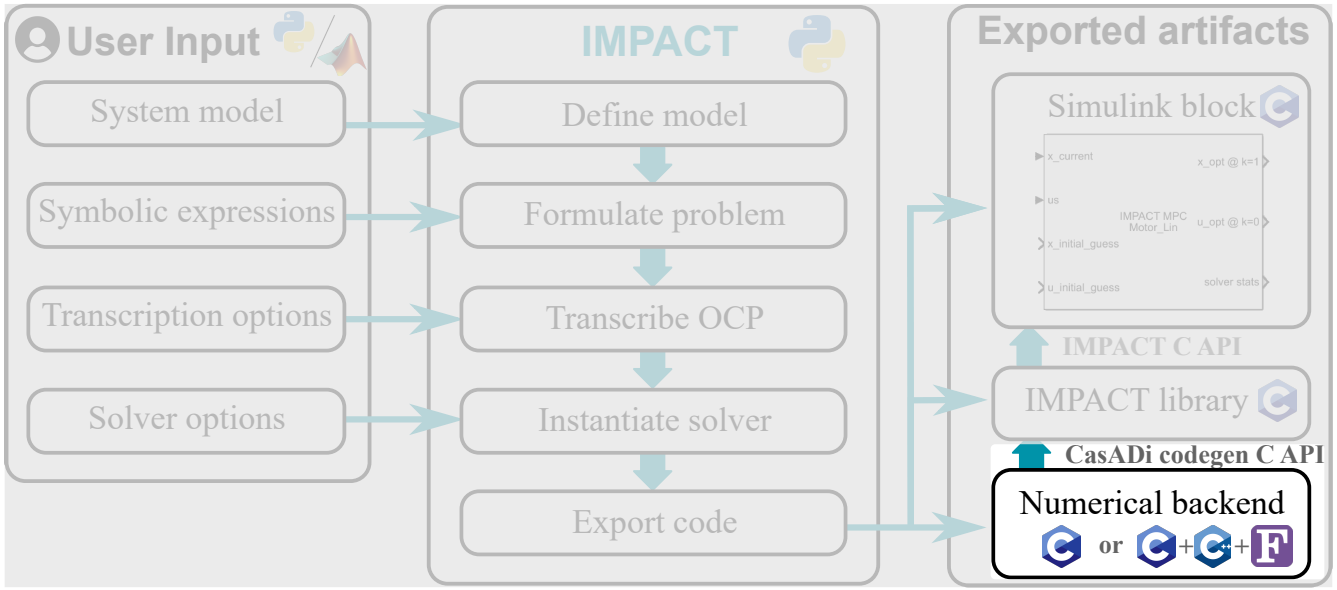
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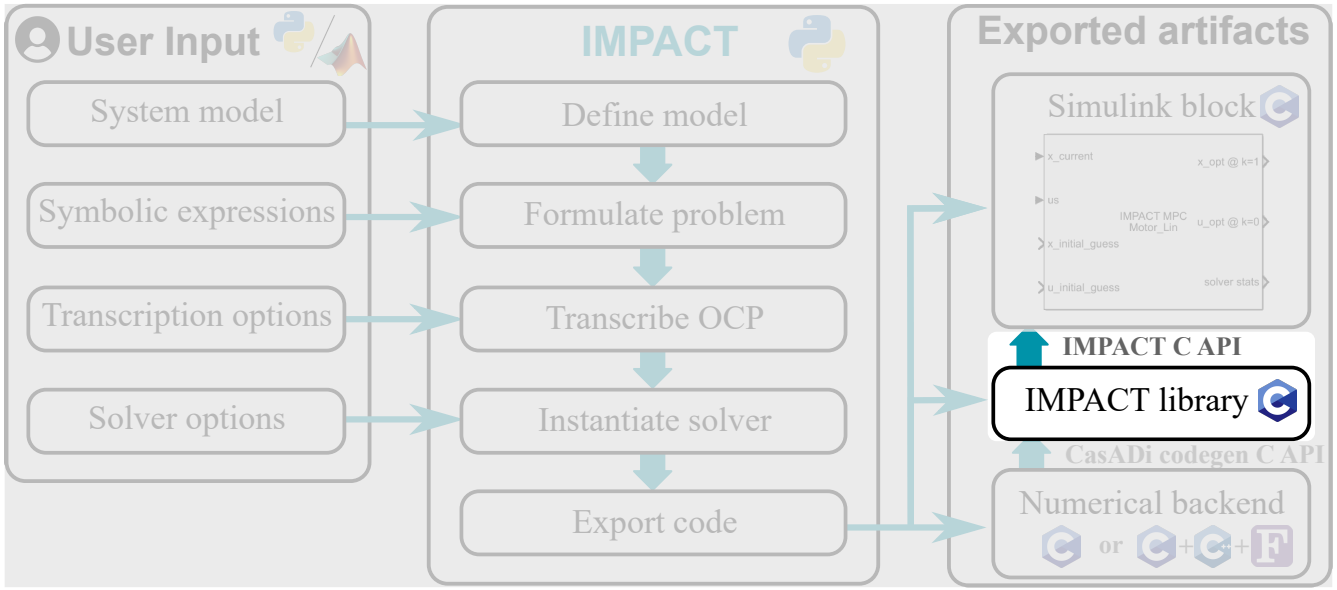
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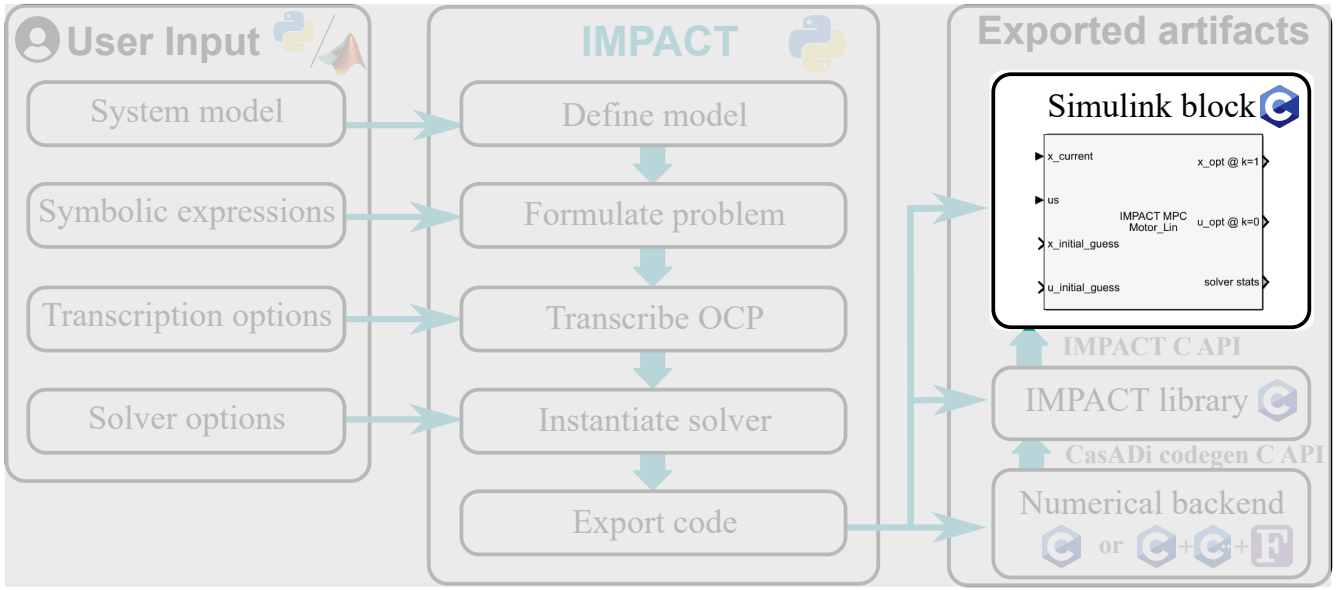
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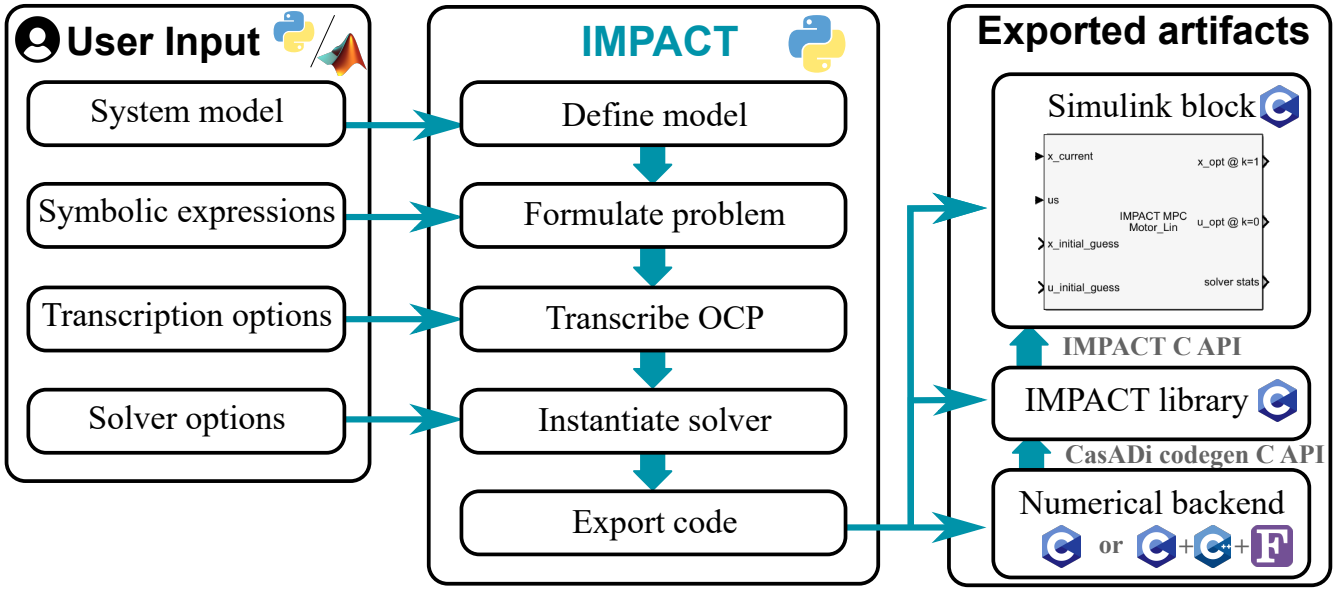
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- The artifacts are self-contained C code with C solver, C++ code that still depends on CasADi and/or third-party solvers.
- The artifacts can be used at different levels for deployment in a wide variety of devices.

More Information

Workshop Today!!!





Thank you for your attention